



Measurement of the calpain1-generated α -synuclein fragment in serum from PPMI cohort subjects.

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Summary

60 serum samples from the PPMI cohort were measured for the presence of a calpain1-generated fragment of α -synuclein to determine if the fragment was able to select individuals diagnosed with Parkinson's Disease from matched controls.

Method

The α -SYN-C assay is a competitive ELISA that measures a neoepitope fragment located in the C-terminal region of α -Synuclein, a sequence which is conserved 100% across humans, mouse, rat, and bovine. The monoclonal antibodies were screened for reactivity towards the standard peptide (synthetic peptide for calibration curve) and native material (human serum). The clone NBH-499-1 was chosen for assay development and determined as an IgG1 subtype. To evaluate the specificity of the α -SYN-C assay, the mAb was tested towards the elongated peptide, truncated peptide, non-sense standard peptide, non-sense coater, and full-length α -Synuclein protein, and showed no reactivity towards those peptides (Figure 1). In vitro cleavage experiments confirmed that cleavage of α -Synuclein by Calpain-1 resulted in the generation of the α -SYN-C fragments within 1 h of cleavage after which they decreased. No reactivity was found towards the non-cleaved α -Synuclein or Calpain-1 confirming very high specificity of the assay for the cleavage epitope. Technical validation was performed to evaluate the novel α -SYN-C assay. Overall, the α -SYN-C assay demonstrated good technical performance (Table [1](#)), including accepted inter- and intra-variations, analyte stability, and no interference from hemoglobin, lipemia and biotin.





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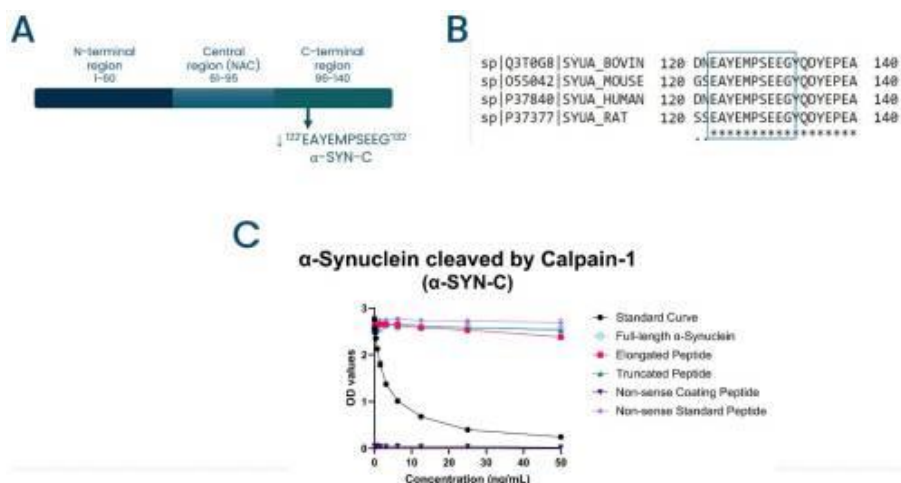


Table 1.
Summary of technical parameters for α -SYN-C.

Technical validation

IC50

Measurement range (LLMR-ULMR)

Inter-assay variation, mean (range)

Intra-assay variation, mean (range)

Dilution recovery of human serum, mean (range)

Spiking recovery (selection peptide in serum), mean (range)

Analyte stability and recovery, 48 h at 4 °C/20°C, mean (range)

Freeze-thaw stability, five cycles, mean (range)

Hemoglobin interference (low/high), mean (range)

Lipemia interference (low/high), mean (range)

Biotin interference

**Results (Accepted recovery:
100 $\pm$ 20%)**

3.44 ng/mL

0.27–30.30 ng/mL

5.8% (2.7–9.6)

12.8% (4.6–21.0)

105% (87.0–106)

105% (88.0–122.0)

92.8% (86.2–100.4) / 92.8% (81.9–102.9)

81.7% (68.0–89.9)

101.3% (100.0–107.4) / 95.1% (88.3–101.6)

106.5% (100.9–108.0) / 110.4% (108.0–113.3)

< 100 ng/mL



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References

Signe Holm Nielsen , Cecilie Møller Hausgaard, Meryem Benmarce, Morten Karsdal, Kim Henriksen. A fragment of Calpain-1 cleaved α -Synuclein quantified in serum is upregulated in patients with Parkinson's disease
Sci Rep . 2025 Apr 9;15(1):12081.

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